

Joint Stiffness–Damping Spectral Geometry for Inverter-Dominated Power Grids: Coupling Corrections, Planning Sensitivities, and Directed Fragility

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Abstract—Inverter-based resources change not only aggregate inertia but also the spatial geometry of damping and controller-mediated disturbance propagation. This paper develops a calibrated spectral framework that separates three quantities that are often conflated: inertia-weighted electrical stiffness, modal damping, and worst-case input–output amplification. For the reduced network model $M\ddot{\theta} + D\dot{\theta} + L\theta = p$, the relevant second-order operator is the quadratic eigenvalue problem (QEP) $s^2I + s\tilde{D} + \tilde{L}$, where $\tilde{L} = M^{-1/2}LM^{-1/2}$ and $\tilde{D} = M^{-1/2}DM^{-1/2}$. When the two normalized operators commute, graph modes decouple exactly and a target damping ratio is the region $\delta_k \geq 2\zeta_*\sqrt{\nu_k}$. For non-proportional damping, we derive a perturbation result showing that purely off-diagonal modal damping produces no first-order pole displacement; the leading correction is second order and depends on both coupling strength and modal separation. This yields a pairwise-coupling correction and a dimensionless diagnostic for deciding when a diagonal modal screen is unreliable. Across 1,100 synthetic networks, the diagnostic attains an AUC of 0.918 for detecting diagonal-margin errors above 2%, while the second-order correction reduces median error from 4.84% to 1.43% under strong heterogeneity. We also derive a coordinate-invariant QEP sensitivity that treats inertia, damping, and line-stiffness changes in one formula; finite differences validate pole and damping-ratio derivatives to median relative errors of 1.47×10^{-6} and 1.28×10^{-6} . A topology-grounded IEEE 39-bus reduced surrogate shows that Fiedler and inertia conversion orders can incur a damping-margin regret of 0.00844 at 80% conversion, while line-stiffness and damping-margin rankings have Kendall $\tau = -0.436$. Finally, a transparent three-node non-reciprocal synchronization-loop example shows distinct worst forcing and response buses and opposite rankings for stiffness reinforcement and resolvent reduction. The result is a metric-dependent definition of dynamic fragility: there is no universal weak bus, weak line, or critical mode independent of the performance quantity being optimized.

Index Terms—Inverter-based resources, quadratic eigenvalue problem, non-proportional damping, spectral graph theory, resolvent analysis, small-signal stability, non-normal systems, power-system planning.

I. INTRODUCTION

THE transition from synchronous generation to inverter-based resources (IBRs) changes several dynamic mechanisms at once. Physical inertia is displaced or emulated, damping becomes controller-dependent, and grid-following or grid-forming synchronization loops introduce additional states and directional couplings. Treating this transition as a change

in one scalar “system inertia” therefore discards the spatial information that determines inter-area oscillations and planning sensitivities.

Graph-spectral descriptions provide a useful bridge between network physics and dynamics because the active-power angle Jacobian is Laplacian-like. However, a graph eigenvalue is a stiffness or frequency coordinate, not a universal stability margin. This distinction is especially important after mass normalization. If ν_k is an eigenvalue of $M^{-1/2}LM^{-1/2}$, then reducing inertia at a participating node can *increase* ν_k . Hence a larger normalized graph eigenvalue cannot automatically mean a safer grid. The damping ratio depends simultaneously on decay and frequency; under proportional damping it scales as $1/\sqrt{\nu_k}$ rather than monotonically with ν_k .

A second limitation appears when damping is non-proportional. The eigenvectors of the normalized stiffness $\tilde{L}$ need not diagonalize the normalized damping $\tilde{D}$. The graph modes then couple through $Q^T \tilde{D} Q$, and a mode-by-mode damping formula loses the off-diagonal information. This is the classical non-proportional-damping issue in a power-network setting, naturally expressed as a QEP [3], [4].

A third distinction concerns poles versus disturbance amplification. Even after the small-signal poles are known, a planner may need to know where a disturbance is most damaging and where the largest response appears. These questions are input–output questions. In a non-normal or non-reciprocal realization, they are described by the resolvent and its singular vectors, not by a scalar Laplacian centrality [11].

Recent work has developed performance metrics for low-inertia networks [5], [6], optimal inertia and primary-control placement [7], decentralized converter stability certificates [8], damping-oriented conditions for heterogeneous GFM/GFL systems [9], and stability manifolds for converter-dominated grids [10]. These results make broad claims that a new graph eigenvalue alone “certifies” an IBR grid difficult to defend. The contribution pursued here is narrower: identify exactly what the stiffness spectrum captures, derive the leading correction when damping destroys modal decoupling, and connect those quantities to planning and input–output fragility without replacing full device-level validation.

The paper makes four contributions.

- 1) It establishes a *joint stiffness–damping geometry*. The region $\delta_k \geq 2\zeta_*\sqrt{\nu_k}$ is exact only when $[\tilde{L}, \tilde{D}] = 0$,

and the paper explicitly separates that statement from non-proportional cases.

- 2) It derives a *second-order modal-coupling correction*. Purely off-diagonal damping does not shift a simple decoupled QEP pole at first order; the leading shift is quadratic in the off-diagonal modal damping. The resulting expression provides both an inexpensive corrected margin and a derived coupling indicator.
- 3) It derives and numerically validates a *generalized QEP sensitivity* for simultaneous changes in M , D , and L . This avoids the coordinate ambiguity that arises when inertia itself changes under mass normalization.
- 4) It shows that planning rankings are metric-dependent. Synthetic ensembles, a reduced IEEE 39-bus topology-grounded study, and a transparent non-reciprocal three-node example demonstrate that stiffness, damping margin, and resolvent amplification can rank buses and lines differently.

The paper does *not* claim that a second-order swing surrogate is a complete high-IBR model. The IEEE 39-bus study reported here is a topology-grounded reduced experiment, not a substitute for a full dq-domain or ANDES/EMT benchmark. This limitation is retained in the main manuscript because it defines the evidence boundary.

II. NETWORK MODEL AND TWO NORMALIZED GEOMETRIES

Consider a connected network linearized around an operating point,

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p, \quad (1)$$

where $M = \text{diag}(M_i) \succ 0$, $D = \text{diag}(D_i) \succeq 0$, and $L = L^T \succeq 0$ is the angle-to-active-power Jacobian under the reduced electromechanical approximation. For a lossless branch (i, j) ,

$$w_{ij} = V_i V_j b_{ij} \cos(\theta_i^* - \theta_j^*), \quad (2)$$

with L assembled as the corresponding weighted Laplacian.

The uniform-angle mode is removed by working on the relative-angle subspace. With the mass coordinate $\vartheta = M^{1/2}\theta$,

$$\ddot{\vartheta} + \tilde{D}\dot{\vartheta} + \tilde{L}\vartheta = M^{-1/2}p, \quad (3)$$

where

$$\tilde{L} = M^{-1/2}LM^{-1/2}, \quad \tilde{D} = M^{-1/2}DM^{-1/2}. \quad (4)$$

Let

$$\tilde{L}Q = Q\Lambda, \quad \Lambda = \text{diag}(\nu_2, \dots, \nu_n), \quad (5)$$

be the eigendecomposition on the nontrivial subspace. In graph-modal coordinates $x = Q^T\vartheta$,

$$\ddot{x} + \Gamma\dot{x} + \Lambda x = Q^T M^{-1/2}p, \quad \Gamma = Q^T \tilde{D}Q. \quad (6)$$

The diagonal terms $\delta_k = \Gamma_{kk}$ are modal damping coefficients and the off-diagonal terms $\Gamma_{k\ell}$ couple graph modes.

A. What the stiffness spectrum does and does not mean

For a simple eigenpair (ν_k, q_k) of $\tilde{L}$, symmetric perturbation gives

$$\frac{\partial \nu_k}{\partial M_i} = -\frac{\nu_k}{M_i} q_{k,i}^2. \quad (7)$$

Thus decreasing M_i tends locally to *increase* ν_k . The normalized spectrum is a frequency/stiffness coordinate, not a monotone “more inertia is larger ν_2 ” safety score.

For an edge $e = (i, j)$ with weight w_e ,

$$\frac{\partial \nu_k}{\partial w_e} = \left(\frac{q_{k,i}}{\sqrt{M_i}} - \frac{q_{k,j}}{\sqrt{M_j}} \right)^2 \geq 0. \quad (8)$$

Equation (8) is an exact stiffness sensitivity for a simple mode. It is not a damping-ratio sensitivity.

III. EXACT DAMPING REGION AND ITS BOUNDARY OF VALIDITY

The exact second-order poles are the roots of

$$P(s)v = (s^2 I + s\tilde{D} + \tilde{L})v = 0. \quad (9)$$

Theorem 1 (Exact commuting damping region). *Suppose $\tilde{L}$ and $\tilde{D}$ are real symmetric on the relative-angle subspace. They decouple into independent scalar QEP modes if and only if they commute,*

$$[\tilde{L}, \tilde{D}] = 0. \quad (10)$$

For each underdamped mode,

$$s_{k,\pm} = -\frac{\delta_k}{2} \pm j\sqrt{\nu_k - \frac{\delta_k^2}{4}}, \quad \zeta_k = \frac{\delta_k}{2\sqrt{\nu_k}}. \quad (11)$$

Therefore a target ζ_* is satisfied if and only if

$$\delta_k \geq 2\zeta_*\sqrt{\nu_k}, \quad k = 2, \dots, n. \quad (12)$$

Proof. Real symmetric matrices are simultaneously orthogonally diagonalizable exactly when they commute. Substitution of the shared eigenbasis into (9) yields the scalar polynomial $s^2 + \delta_k s + \nu_k = 0$. Equation (11) then gives (12). $\square$

Figure 1 illustrates the region. Its role is deliberately calibrated: it is an exact certificate in the commuting case and a diagonal screen otherwise.

A useful immediate consequence is that the Fiedler mode need not be damping-critical. If $\tilde{D} = \gamma I$, then $\zeta_k = \gamma/(2\sqrt{\nu_k})$; among underdamped modes the *largest* ν_k , not ν_2 , gives the smallest damping ratio.

IV. SECOND-ORDER CORRECTION FOR NON-PROPORTIONAL DAMPING

The principal mathematical addition in this version is an explicit perturbation formula for the information missing from the diagonal screen.

Write the modal damping as

$$\Gamma = \Gamma_d + \epsilon\Gamma_o, \quad (13)$$

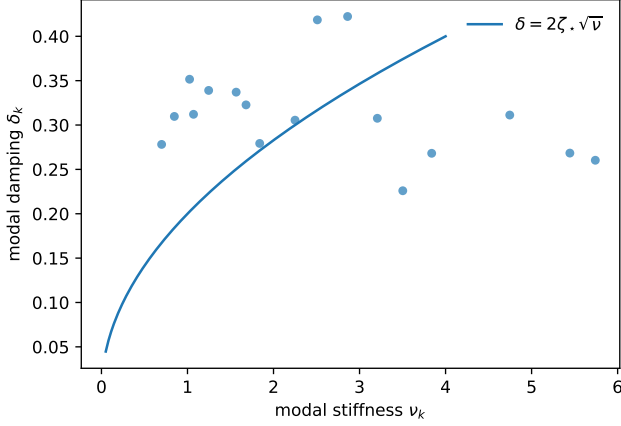


Fig. 1. Damping-region coordinates. The boundary $\delta = 2\zeta_* \sqrt{\nu}$ is exact for commuting stiffness and damping and becomes a diagonal screen under non-proportional damping.

where $\Gamma_d = \text{diag}(\delta_k)$ and Γ_o has zero diagonal. Define

$$p_k(s) = s^2 + \delta_k s + \nu_k. \quad (14)$$

Let $s_k^{(0)}$ be a simple root of $p_k(s) = 0$.

Theorem 2 (Leading pole shift from off-diagonal modal damping). *For a simple, separated decoupled root $s_k^{(0)}$, the perturbed QEP pole admits*

$$s_k(\epsilon) = s_k^{(0)} + \epsilon^2 s_k^{(2)} + \mathcal{O}(\epsilon^3), \quad (15)$$

with zero first-order shift and

$$s_k^{(2)} = \frac{(s_k^{(0)})^2}{2s_k^{(0)} + \delta_k} \sum_{\ell \neq k} \frac{\Gamma_{k\ell}^2}{p_\ell(s_k^{(0)})}. \quad (16)$$

Proof. Let $v = e_k + \epsilon v_1 + \epsilon^2 v_2 + \dots$ and $s = s_k^{(0)} + \epsilon s_1 + \epsilon^2 s_2 + \dots$. At first order, projection on e_k gives $(2s_k^{(0)} + \delta_k)s_1 + s_k^{(0)}\Gamma_{kk}^o = 0$. Because $\Gamma_{kk}^o = 0$, $s_1 = 0$. For $\ell \neq k$, the first-order eigenvector coefficient is

$$(v_1)_\ell = -\frac{s_k^{(0)}\Gamma_{\ell k}}{p_\ell(s_k^{(0)})}. \quad (17)$$

Projecting the second-order equation on e_k yields

$$(2s_k^{(0)} + \delta_k)s_2 + s_k^{(0)}(\Gamma_o v_1)_k = 0, \quad (18)$$

which gives (16) using the symmetry of Γ . $\square$

Equation (16) explains two empirical regimes. If modal roots are separated, $|p_\ell(s_k^{(0)})|$ is not small and the diagonal approximation is accurate to first order even with visible heterogeneity. Near degeneracy, the same off-diagonal $\Gamma_{k\ell}$ is divided by a small modal denominator and the diagonal mode loses meaning.

We define the pairwise-corrected pole

$$\hat{s}_k^{\text{PSC}} = s_k^{(0)} + s_k^{(2)} \quad (19)$$

and a dimensionless, theory-derived warning index

$$\eta = \max_k \left| \frac{s_k^{(2)}}{s_k^{(0)}} \right|. \quad (20)$$

Unlike a raw commutator norm, η weights coupling by the actual QEP pole separation. The normalized commutator

$$\mathcal{C} = \frac{\|\tilde{L}, \tilde{D}\|_F}{\|\tilde{L}\|_F \|\tilde{D}\|_F} \quad (21)$$

remains useful as a simultaneous-diagonalization diagnostic, but it is not equivalent to non-normality of a first-order realization.

A. Adaptive screening logic

The practical hierarchy is:

- 1) compute all diagonal modal damping ratios from (11);
- 2) compute the PSC correction (16) and η ;
- 3) accept the corrected/diagonal screen when η is small;
- 4) otherwise solve a reduced QEP containing the lowest estimated-damping modes and their strongly coupled neighbors;
- 5) use the full eigensolve whenever reduced-order assumptions are not trustworthy.

The reduced QEP is therefore a safeguard, not a claim that full eigenanalysis is intrinsically expensive for every network.

V. COORDINATE-INVARIANT SENSITIVITY OF THE PHYSICAL QEP

Mass-normalized sensitivities are convenient when M is fixed. They become awkward when inertia itself is the planning variable because both $\tilde{L}$ and $\tilde{D}$ change with M . It is cleaner to differentiate the physical-coordinate QEP

$$\mathcal{P}(s, \rho)v = (s^2 M(\rho) + sD(\rho) + L(\rho))v = 0. \quad (22)$$

Theorem 3 (Generalized QEP pole sensitivity). *Let $s(\rho)$ be a simple QEP eigenvalue, with right vector v and left vector y satisfying $\mathcal{P}(s, \rho)v = 0$ and $y^H \mathcal{P}(s, \rho) = 0$. Then*

$$\frac{ds}{d\rho} = -\frac{y^H (s^2 M_\rho + sD_\rho + L_\rho) v}{y^H (2sM + D) v}. \quad (23)$$

Proof. Differentiate $\mathcal{P}(s(\rho), \rho)v(\rho) = 0$ and premultiply by y^H . The term containing $dv/d\rho$ vanishes because $y^H \mathcal{P} = 0$. Collecting the terms proportional to $ds/d\rho$ gives (23). $\square$

Equation (23) contains line, damping, and inertia perturbations in the same expression. For example,

$$L_{w_e} = (e_i - e_j)(e_i - e_j)^T, \quad (24)$$

$$D_{D_i} = e_i e_i^T, \quad M_{M_i} = e_i e_i^T. \quad (25)$$

If $s = \sigma + j\omega$ and $ds/d\rho = a + jb$, the damping ratio $\zeta = -\sigma/|s|$ obeys

$$\frac{d\zeta}{d\rho} = \frac{\sigma\omega b - \omega^2 a}{|s|^3}. \quad (26)$$

This is the correct local quantity for damping-aware reinforcement when the critical pole remains simple and does not switch identity.

VI. INPUT-OUTPUT FRAGILITY AND DIRECTED WEAKNESS

The QEP determines modal poles. It does not by itself determine the worst disturbance shape. Let a first-order realization be

$$\dot{x} = Ax + Bp, \quad y = Cx. \quad (27)$$

Its forced frequency response is

$$H(j\Omega) = C(j\Omega I - A)^{-1}B, \quad (28)$$

with peak gain

$$g_{\max} = \sup_{\Omega} \sigma_{\max}(H(j\Omega)). \quad (29)$$

At a maximizing frequency, let

$$H(j\Omega^*) = U\Sigma V^H. \quad (30)$$

The leading right singular vector $\psi = V(:, 1)$ is the worst forcing direction; the leading left singular vector $\phi = U(:, 1)$ is the corresponding response direction. A directed source-response score is

$$W_{i \rightarrow j} = |\psi_i|^2 |\phi_j|^2. \quad (31)$$

In general $W_{i \rightarrow j} \neq W_{j \rightarrow i}$.

Equation (29) is a forced input-output quantity. It should not be confused with arbitrary initial-condition transient growth, which would be evaluated through $\sup_t \|Ce^{At}B_0\|_2$.

VII. METRIC-STRATIFIED DYNAMIC HOSTING

The corrected framework replaces a single-eigenvalue “hosting polytope” by an intersection of explicit performance constraints. For penetration/controller parameter vector ρ , define

$$\mathcal{H}_{\zeta} = \{\rho : \zeta_{\min}(\rho) \geq \zeta_{\star}\}, \quad (32)$$

$$\mathcal{H}_g = \{\rho : g_{\max}(\rho) \leq g_{\star}\}, \quad (33)$$

$$\mathcal{H}_{\text{op}} = \{\rho : h_{\text{op}}(\rho) \leq 0\}, \quad (34)$$

where h_{op} collects voltage, current, angle, and device constraints. Then

$$\mathcal{H}_{\text{dyn}} = \mathcal{H}_{\zeta} \cap \mathcal{H}_g \cap \mathcal{H}_{\text{op}}. \quad (35)$$

A local tangent budget can be obtained from any smooth margin $m(\rho)$ through $m(\rho_0) + \nabla m(\rho_0)^T(\rho - \rho_0) \geq 0$, but it is only a local first-order certificate unless curvature or remainder bounds are supplied.

VIII. NUMERICAL EXPERIMENTS

All experiments use fixed random seed 20260822. The scripts used to generate every reported table and figure are included with the artifact package. The full QEP eigensolve is always the numerical reference for the reduced second-order model.

TABLE I
SYNTHETIC RELATIVE ERROR VERSUS FULL QEP (%).

Regime	Diagonal		PSC		Reduced QEP	
	Med.	P95	Med.	P95	Med.	P95
Proportional	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0
Weak	0.016	0.312	0.0034	0.064	0.0034	0.030
Moderate	0.485	6.685	0.115	2.929	0.114	1.049
Strong	4.843	30.265	1.432	25.877	1.355	7.799
Clustered	19.785	89.892	17.797	87.526	2.985	32.176

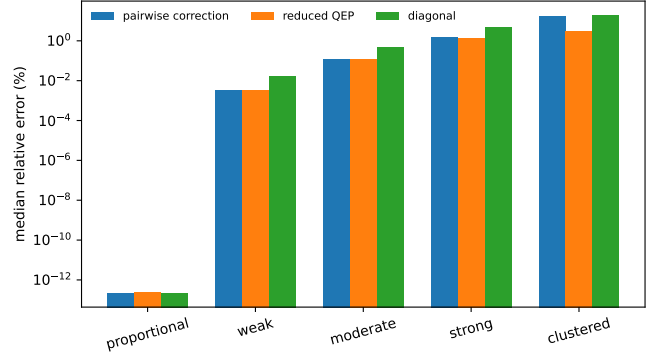


Fig. 2. Median error across 1,100 synthetic networks. The PSC captures the leading non-proportional correction when modes remain separated; the reduced QEP is required in the clustered regime.

A. Synthetic coupling ensembles

We generated 220 connected $n = 24$ networks in each of five regimes: proportional damping, weak heterogeneity, moderate heterogeneity, strong heterogeneity, and a deliberately near-degenerate clustered spectrum. Inertias are heterogeneous; the Laplacian is scaled so the median nonzero modal frequency is 1.5 rad/s. Damping-per-inertia ratios are lognormally perturbed around a common baseline. The clustered ensemble uses an almost-complete graph with small edge perturbations to create nearly repeated stiffness modes.

Table I compares the diagonal all-mode screen, the pairwise second-order correction (PSC), and a seven-mode damping/coupling-selected reduced QEP against the full $2n$ eigensolve.

Two findings matter. First, the absence of a first-order pole shift is visible numerically: weak heterogeneity produces very small diagonal errors, and the PSC lowers them by approximately a factor of five. Second, the PSC is not a cure for degeneracy; its perturbation denominator predicts its own failure. The reduced QEP is therefore retained for the clustered regime.

The derived indicator η tracks diagonal-screen failure closely. Across all cases, the Spearman correlation between $\log \eta$ and $\log$ diagonal error is 0.875. Treating a diagonal error above 2% as “failure”, η obtains an AUC of 0.918. A threshold fitted only on a 60% calibration split ($\eta = 0.0238$) yields precision 0.743, recall 0.897, and specificity 0.797 on the held-out 40%. The threshold is experiment-specific and is not proposed as a universal grid constant.

B. Validation of the generalized QEP sensitivity

We next generated 180 independent $n = 12$ heterogeneous networks. In equal thirds of the runs, the perturbed parameter

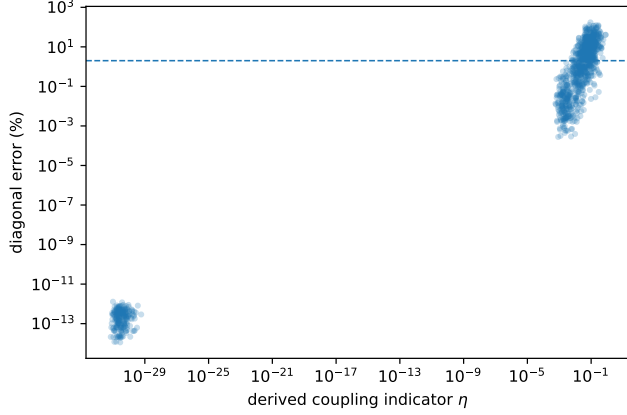


Fig. 3. Diagonal damping-margin error versus the second-order coupling indicator η . The dashed line marks the 2% failure level used only for the diagnostic experiment.

TABLE II
RELATIVE SENSITIVITY ERROR VERSUS FINITE DIFFERENCES.

Parameter	$ds/d\rho$		$d\zeta/d\rho$	
	Median	P95	Median	P95
Damping	9.77×10^{-7}	8.28×10^{-5}	2.85×10^{-7}	1.04×10^{-5}
Edge	1.12×10^{-6}	8.09×10^{-5}	1.22×10^{-6}	4.97×10^{-5}
Inertia	2.40×10^{-6}	5.56×10^{-5}	3.31×10^{-6}	4.19×10^{-5}

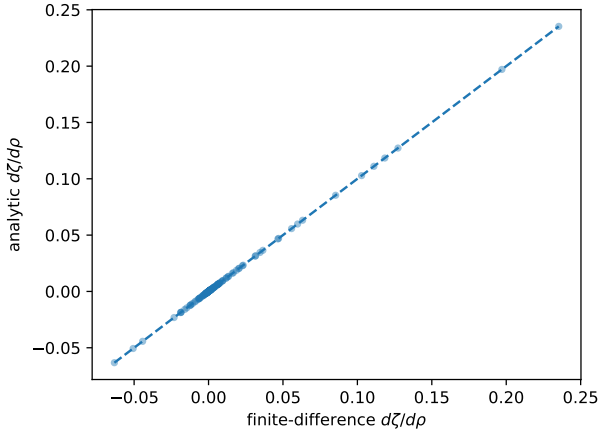


Fig. 4. Analytic versus finite-difference damping-ratio derivatives over edge, damping, and inertia perturbations.

was an edge stiffness, a nodal damping coefficient, or a nodal inertia. The analytic derivative (23) was compared with a 10^{-6} -scale finite difference, matching the same simple pole after perturbation.

Across all parameter classes, the median relative error in $ds/d\rho$ is 1.47×10^{-6} and the 95th percentile is 7.25×10^{-5} . For $d\zeta/d\rho$, the corresponding errors are 1.28×10^{-6} and 3.10×10^{-5} . Table II reports the class-wise results.

This experiment also clarifies the role of the generalized formula: differentiating only $\tilde{L}$ is sufficient for a line change with fixed M , but not for a finite SG-to-IBR conversion in which inertia changes the normalization itself.

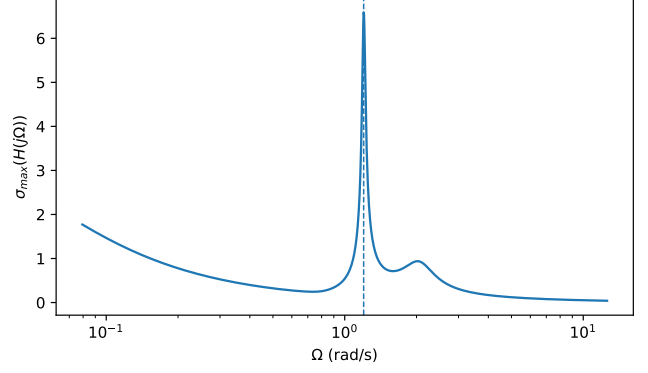


Fig. 5. Resolvent gain of the transparent three-node non-reciprocal example. The maximum occurs near 1.20 rad/s.

C. A transparent non-reciprocal three-node mechanism

To isolate input–output directionality, consider a three-node network with

$$w_{12} = 5, \quad w_{23} = 4, \quad w_{13} = 3, \quad (36)$$

$$M = \text{diag}(10, 2, 6), \quad D = \text{diag}(0.4, 1.2, 0.5). \quad (37)$$

In normalized coordinates, we introduce one explicitly stated one-sided kinematic coupling from normalized frequency at node 3 to the angle-rate state at node 2. With $z = [\vartheta^T \dot{\vartheta}^T]^T$, the first-order realization is

$$A_\kappa = \begin{bmatrix} 0 & I + \kappa e_2 e_3^T \\ -\tilde{L} & -\tilde{D} \end{bmatrix}, \quad \kappa = 0.7. \quad (38)$$

This is a minimal non-reciprocal synchronization-loop surrogate, not an asserted detailed PLL model. Inputs are nodal active-power perturbations and outputs are physical frequency deviations, with $B = [0 \ M^{-1/2}]^T$ and $C = [0 \ M^{-1/2}]$.

The system has $\nu_2 = 1.388$ and all nontrivial poles remain stable with spectral abscissa -0.0202 . Yet the resolvent peaks at

$$g_{\max} = 6.59, \quad \Omega^* = 1.20 \text{ rad/s}. \quad (39)$$

The worst forcing energy is concentrated at nodes $(N_1, N_2, N_3) = (0.254, 0.022, 0.724)$, while the response energy is $(0.046, 0.625, 0.329)$. The dominant forcing node is therefore N_3 , whereas the dominant response node is N_2 .

Planning rankings also diverge. Table III shows finite actions. All three line reinforcements increase ν_2 . Reinforcing line 1–2 reduces $g_{\max}$ by 3.25, while reinforcing line 2–3 increases it by 44.6 and moves the dominant pole close to the imaginary axis. The line with the largest stiffness increase, 1–3, also worsens the resolvent by 5.02. Likewise, adding the same $+0.5$ local damping at node 3 is most beneficial, whereas adding it at node 2 worsens the resolvent in this non-reciprocal realization.

This example should not be labeled a new Braess paradox. Braess-like phenomena in oscillator and power networks are already established [12], [13]. The verified statement is more precise: a stiffness-improving action need not improve either damping ratio or the chosen input–output gain.

TABLE III
THREE-NODE FINITE-ACTION COMPARISON.

Action	$\Delta\nu_2$	$\Delta g_{\max}$	spectral abscissa
$w_{12} + 2$	+0.090	-3.254	-0.0386
$w_{23} + 2$	+0.112	+44.583	-0.00254
$w_{13} + 2$	+0.531	+5.015	-0.0130
$D_1 + 0.5$	-	-1.657	-0.0270
$D_2 + 0.5$	-	+2.608	-0.0138
$D_3 + 0.5$	-	-4.044	-0.0528

D. IEEE 39-bus topology-grounded reduced experiment

The final experiment uses the solved MATPOWER case39 topology [14]. The active-angle surrogate weight for branch $e = (i, j)$ is

$$w_e = \frac{V_i V_j}{|x_e| t_e} \cos(\theta_i^* - \theta_j^* - \phi_e), \quad (40)$$

where $t_e = 1$ for a unit-tap branch. The resulting 39-node Laplacian is Kron-reduced to generator buses 30–39. Generator inertia constants and ratings are taken from a public IEEE 39-bus dynamic parameter set [15]; masses are normalized by their median. For a controlled comparison rather than an absolute-frequency reproduction, the reduced stiffness is scaled so the median nonzero modal frequency is 1.3 rad/s.

The baseline normalized damping-per-inertia profile is deliberately heterogeneous,

$$[0.29, 0.37, 0.24, 0.45, 0.21, 0.33, 0.19, 0.41, 0.17, 0.31], \quad (41)$$

so this case is a reproducible *surrogate* for testing the mathematics on a realistic topology; it is not a claim about the actual damping of the New England system. A GFL-like conversion replaces candidate inertia by 20% of its original value and assigns a lower candidate-dependent normalized damping coefficient.

For the ten single-conversion cases, the diagonal approximation is already good: its worst relative damping-margin error is 1.07%, while the PSC worst error is 0.160% and the reduced-QEP worst error is 0.192%. This is an important negative result: the reduced QEP is not needed to headline this particular surrogate. The synthetic clustered study, not the 39-bus surrogate, is where the reduced QEP earns its role as a safeguard.

The planning result is nevertheless strong. A greedy exact-QEP conversion sequence and the diagonal/PSC sequences select the same buses through 80% conversion. In contrast, inertia-only and Fiedler-participation orders end at $\zeta_{\min} = 0.01961$, while the exact-QEP-guided order retains $\zeta_{\min} = 0.02805$, a damping-margin regret of 0.00844.

Line reinforcement gives a second ranking test. Each of the 46 original branches is strengthened by 10%, the 39-node stiffness matrix is rebuilt, and the generator Kron reduction is recomputed. As expected, $\Delta\nu_2 \geq 0$ for all passive stiffness reinforcements. The relationship with damping margin is different: Kendall rank correlation between $\Delta\nu_2$ and $\Delta\zeta_{\min}$ is -0.436 . The 16–17 branch gives the largest $\Delta\nu_2 = 0.00814$ but almost no change in damping margin. In contrast, strengthening 20–34 and 19–20 produces only $\Delta\nu_2 = 0.000404$ and 0.000313 , yet reduces $\zeta_{\min}$ by 0.00161 and 0.00124 , respectively. Thus a positive stiffness sensitivity

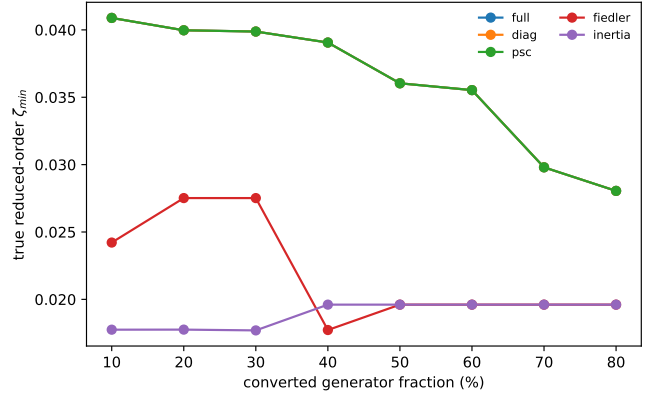


Fig. 6. True reduced-order damping margin under progressive conversion orders on the IEEE 39-bus topology-grounded surrogate. The full-QEP, diagonal, and PSC orders coincide; inertia and Fiedler heuristics do not.

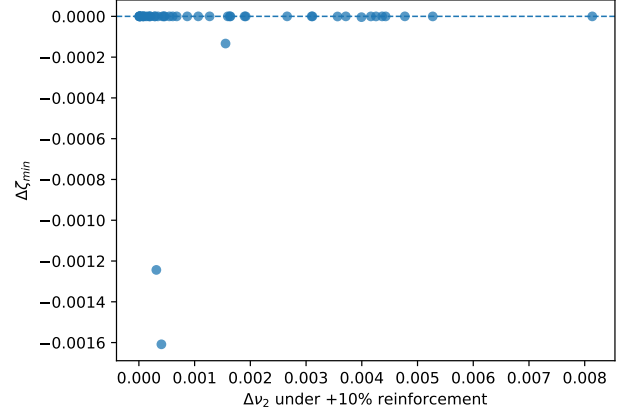


Fig. 7. Line-stiffness improvement versus damping-margin change for +10% reinforcement of each case39 branch. The rankings are negatively correlated ($\tau = -0.436$).

does not determine the sign or importance of the damping-margin change.

E. Computation-time boundary

The numerical goal is calibrated screening, not the claim that a full eigensolve is infeasible. On the reference Python implementation, at $n = 120$ the median times are 29.1 ms for the full $2n$ eigensolve, 1.93 ms for the diagonal screen, and 14.1 ms for the PSC. The current unoptimized mode-selection implementation of the reduced QEP takes 64.4 ms and is therefore *slower* than the full solve at this size. We retain the reduced QEP because it repairs clustered modes, not because this prototype has demonstrated a runtime advantage. A production implementation would require sparse eigensolvers and incremental mode tracking before computational superiority could be claimed.

IX. PLANNING IMPLICATIONS

The experiments support three practical statements.

A. A weak bus is metric-dependent

A bus can be important for $\partial\nu_k/\partial M_i$, another can dominate the least-damped QEP mode, and another can be the leading resolvent forcing or response bus. These are different questions, not contradictory rankings.

For a small damping addition with fixed M and L , the diagonal-regime sensitivity is

$$\frac{\partial\zeta_k}{\partial D_i} \approx \frac{q_{k,i}^2}{2M_i\sqrt{\nu_k}}, \quad (42)$$

but finite actions and strongly coupled modes should be evaluated with (23) or by recomputing the QEP.

B. A weak line is also metric-dependent

Differentiating the diagonal formula gives

$$\frac{\partial\zeta_k}{\partial w_e} = \frac{1}{2\sqrt{\nu_k}} \frac{\partial\delta_k}{\partial w_e} - \frac{\delta_k}{4\nu_k^{3/2}} \frac{\partial\nu_k}{\partial w_e}. \quad (43)$$

The second term is negative whenever reinforcement increases ν_k . The first term contains eigenvector rotation relative to the damping geometry. Equation (43) therefore explains why a monotone stiffness improvement can coexist with a lower damping ratio.

C. No universal critical mode

The Fiedler mode is structurally important for connectivity and low-frequency coherence, but the minimum damping ratio need not occur there. Likewise, the largest resolvent gain can occur at a frequency near a pole while its singular vectors define spatial directions that are not recoverable from one graph eigenvector. Planning should therefore name the objective explicitly: stiffness, damping margin, stochastic variance, forced amplification, or another operational metric.

X. LIMITATIONS AND FALSIFICATION CRITERIA

The strongest limitation is model fidelity. Equation (1) omits voltage/reactive-power states, current-control dynamics, PLL internal states, limiters, line dynamics, and protection logic. The non-reciprocal term in (38) is intentionally a minimal mechanism model and is not presented as a validated PLL realization. The IEEE 39-bus experiment uses the authentic topology and solved operating point, but its damping and conversion models are controlled surrogates.

A full journal empirical claim therefore requires a second validation layer: construct L, M, D from a documented dynamic benchmark, compare the reduced critical modes against the full linearized IBR state matrix, and repeat the conversion and line-ranking experiments with full-model poles. If the scalar surrogate loses pole correspondence at high IBR penetration, the correct interpretation is “screening model with an identified validity region”, not a universal stability certificate.

The present evidence can also falsify parts of the proposed method. If future full-model cases remain weakly coupled, the diagonal all-mode formula should remain the principal tool and the reduced QEP should be demoted to a rare safeguard. If damping-aware and stiffness-only line rankings agree on a

high-fidelity benchmark, the line result becomes a theoretical caution rather than a frequent planning reversal. If the resolvent and modal rankings do not differ after physically validated controller models are introduced, directed fragility should remain a diagnostic extension rather than a headline result.

XI. CONCLUSION

This paper reorganized the spectral analysis of inverter-dominated grids around a simple separation: stiffness, damping, and disturbance amplification are different properties.

The mass-normalized Laplacian $\tilde{L}$ provides the electrical/inertial stiffness geometry. The pair $(\tilde{L}, \tilde{D})$ determines the QEP and damping margin. In the commuting case, the damping region $\delta_k \geq 2\zeta_*\sqrt{\nu_k}$ is exact. Under non-proportional damping, the leading effect of off-diagonal modal damping on a simple separated pole is second order; the derived PSC and indicator η quantify when the diagonal screen is safe and when it should be abandoned. A generalized physical-coordinate QEP sensitivity then treats inertia, damping, and line changes consistently.

The numerical evidence supports the resulting hierarchy. The PSC substantially reduces error under heterogeneous but separated modes, while clustered modes require a coupled QEP. The generalized sensitivity agrees with finite differences across all three physical perturbation classes. On the case39 topology-grounded surrogate, damping-aware conversion planning retains a materially larger margin than Fiedler or inertia orders, and stiffness and damping line rankings disagree. In a transparent non-reciprocal example, the worst forcing bus differs from the response bus and a line that strongly raises ν_2 can worsen the resolvent.

The main conclusion is therefore not that one new eigenvalue replaces simulation. It is that a planning statement is incomplete until the relevant performance geometry is named. A grid has no universal weak bus, weak line, or critical mode independent of whether the objective is modal stiffness, damping margin, or input-output amplification.

APPENDIX A

DETAILED DERIVATION OF THE SECOND-ORDER COUPLING FORMULA

For completeness, write

$$P(s, \epsilon) = P_0(s) + \epsilon s \Gamma_o, \quad (44)$$

where $P_0(s) = s^2 I + s \Gamma_d + \Lambda$. Let $P_0(s_k^{(0)})e_k = 0$ and expand s and v as in (15). The order- ϵ equation is

$$P_0(s_0)v_1 + [(2s_0 I + \Gamma_d)s_1 + s_0 \Gamma_o]e_k = 0. \quad (45)$$

The k th component gives $s_1 = 0$. For $\ell \neq k$,

$$(v_1)_\ell = -s_0 \Gamma_{\ell k} / p_\ell(s_0). \quad (46)$$

At order ϵ^2 , only the terms needed by the k th projection are

$$(2s_0 + \delta_k)s_2 + s_0 \sum_{\ell \neq k} \Gamma_{k\ell}(v_1)_\ell = 0, \quad (47)$$

which yields (16). The expansion is nonuniform as $p_\ell(s_0) \rightarrow 0$, exactly the repeated/clustered-mode case.

APPENDIX B REPRODUCIBILITY DETAILS

The synthetic study uses $n = 24$, 220 realizations per regime, and seed 20260822. The reduced QEP retains seven modes. The QEP-sensitivity test uses 180 $n = 12$ networks with an equal split among edge, damping, and inertia perturbations. Finite differences use a perturbation of 10^{-6} times the relevant parameter scale.

The case39 experiment uses the 46 in-service MATPOWER branches and generator buses 30–39. The public inertia/rating set in [15] is

$$H = [5.00, 4.33, 4.48, 3.58, 4.33, 4.35, 3.77, 3.47, 3.45, 4.00] \text{ s}, \quad (48)$$

with generator ratings, in the same order,

$$[1000, 700, 800, 800, 300, 800, 700, 700, 1000, 1000] \text{ MVA}. \quad (49)$$

The controlled GFL-like normalized damping profile used after conversion is

$$[0.16, 0.22, 0.12, 0.30, 0.14, 0.25, 0.11, 0.28, 0.10, 0.20]. \quad (50)$$

These values define a reproducible reduced-order stress test and should not be interpreted as manufacturer data.

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